

PROFESSIONAL SUMMARY

Trained in human-centered design and computer science, I combine analytical rigor with data storytelling to uncover and communicate insights that shape data-informed product strategies. Drawing from my architectural background in studying user behavior and spatial interaction, I approach data not just as numbers, but as narratives that reveal how people engage with products and technology—transforming analysis into meaningful actions that drive better user experiences.

EDUCATION

University of Southern California	Los Angeles, CA
<i>Master of Science in Computer Science (Artificial Intelligence)</i>	January 2025 — December 2026 (Expected)
DSCI 550: Data Science at Scale - Course Grader (25 fall), 2025 IMSC(Integrated Media Systems Center)	Hackathon Organizer
State University of New York at Oswego	Oswego, NY
<i>Exchange Student</i>	August 2021 — December 2021
Hanyang University	Seoul, Korea
<i>Bachelor of Science in Computer Science & Interior Architecture Design</i> , Summa Cum Laude, GPA: 4.04/4.5	March 2019 — August 2024
Undergraduate Leadership Scholarship, Hanyang Talent Excellence Award Scholarship	

INTERNSHIPS

AIoT Group at Seoul National University <i>Research Member</i>	Seoul, Korea February 2024 — December 2024
- Contributed to a sensor control system optimizing image acquisition under domain shift, enhancing robustness of AI models for field deployment in real-world environments. - Designed evaluation protocols on real-world datasets (ImageNet-ES Natural), yielding performance gains in object recognition. - Led diverse parameter-based dataset collection under strict deadlines, ensuring data diversity for robust model testing.	
Multimodal AI Lab at Hanyang University <i>Undergraduate Researcher</i>	Seoul, Korea August 2023 - January 2024
- Researched multimodal classification integrating vision, geospatial, and text metadata for real-world recognition tasks. - Modified and extended Transformer-based CLIP for contextual awareness, achieving performance gains in fine-grained classification. - Strengthened lab-wide collaboration through weekly presentations on project updates and state-of-the-art vision-language research.	
COGA Robotics <i>Intern Researcher</i>	Seoul, Korea August 2022 — February 2023
- Contributed at the Advanced Technology Research Center, focusing on software and hardware for indoor mobile robots. - Built sensor perception pipelines by integrating LiDAR and RGB-D cameras in ROS, enabling spatial awareness for indoor robots. - Optimized Deep SORT tracker to run in real-time on Jetson Nano, improving tracking accuracy with minimal latency. - Investigated and benchmarked recent SOTA models to propose deep learning-based enhancements for autonomous navigation.	

PUBLICATIONS

Pre- and Post-Occupant Satisfaction in Modern Office Hot-Desking Systems: A Comparative Study of Workplace Environments	Under review
Zhihong Liu, Yifan Zhang, Tsz Him Ian Chu, Ghazal Chizarian, Joon Ho Choi, and Yoomin Lee	

SYMPOSIUM PRESENTATIONS

Toward Robust and Scalable Multi-View Truck Detection via Semi-Supervised Learning	Las Vegas, NV
Poster presentation at The 25th KOCSEA Technical Symposium 2025	November 2025
Built a multi-view object detection system (YOLOv8, PyTorch) for transportation monitoring, using a teacher→student pseudo-labeling pipeline for domain-robust deployment across camera angles.	

AWARDS

SNU Ambient AI Competition (1st Prize) — Seoul National University	Seoul, Korea
Created on-device real-time blurring application. Team of 4 (Link: Youtube, Github)	September 2024
DB GAPS Challenge (2nd Place) — DB Kim Jun Ki Cultural Foundation	Seoul, Korea
Used financial and market data to build automated investment strategy with team of 4. (Link)	November 2020

SELECTED PROJECTS

Tracking: Person Following for Indoor Robots with Camera — <i>Python, ROS</i>	December 2022 — February 2023
- Built a ROS tracking pipeline fusing RGB-D and LiDAR; integrated Deep SORT and sensor calibration for stable trajectories. - Prepared a real-time deployment with modular nodes, launch files, and monitoring, validating robustness through failure analysis.	
OLAP vs. OLTP DBMS Analysis — <i>DuckDB, PostgreSQL, C++, System Internals</i>	November 2024 — December 2024
- Compared DuckDB and PostgreSQL internals (scan ops, buffer mgmt, file formats) to assess suitability for ML data pipelines. - Benchmarked sequential query performance and analyzed trade-offs for analytical workloads relevant to feature engineering. - Derived practical guidelines for DBMS selection and gained hands-on experience with SQL optimization for scalable data prep.	
Fine-grained Classification with Additional Language Data — <i>Python, TensorFlow</i>	October 2023 — January 2024
- Integrated spatio-temporal priors and CLIP functions to enhance baseline performance. - Reproduced and modified multiple models to evaluate additional information for fine-grained classification.	
Classifying Similar Furniture — <i>Python, TensorFlow</i>	March 2023 — June 2023
- Bridged design and ML to classify furniture by generating a class-balanced image set from 3D models with multi-view and light renders. - Built an end-to-end pipeline—from dataset spec and splits to TensorFlow training/eval—showcasing involvement from data prep to delivery.	

SKILLS

Language: C/C++, Python, Matlab, JAVA, PostgreSQL, L^AT_EX Library: PyTorch, TensorFlow, Numpy, OpenCV Frameworks & Tools: ROS, Git, Linux, Dockers Graphics: Autodesk AutoCAD, Figma, Adobe Photoshop, Adobe Illustrator, Adobe InDesign, Rhino/Grasshopper	
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RELATED COURSES

- **Core:** Probability & Statistics / System Programming / Data Structures / Database Systems / Algorithms and Problem Solving / Computer Architecture / Micro-Processor Applications / Object-Oriented Systems Design / High Performance Computing and Simulations
- **Vision & AI:** Machine Learning / Computer Vision / Deep Learning Methods and Applications / Data Science / Computer Science Capstone PBL / Computer Graphics / Design Computing / VR AR Design / Creative Computing